BCSE0593: DATA SCIENCE AND VISUALIZATION LAB

Experiment 01: Exploring Kaggle & GitHub

Create accounts, download a dataset from Kaggle, and upload analysis to GitHub repository

Experiment No: 01
Tools: Kaggle, GitHub

1. LEARNING OBJECTIVES

- Set up user accounts on platforms: Kaggle and GitHub.
- Search, filter, and extract open-source datasets from Kaggle.
- Perform a basic preliminary dataset analysis using Python.
- Initialize a remote GitHub repository and push code/analysis files.

2. SOFTWARE & PREREQUISITES

- **Datasets / Tools:** Kaggle Platform, GitHub Web/Git CLI
- **Environment:** Python 3.x, Jupyter Notebook / Google Colab / VS Code
- **Browser:** Modern Web Browser (Chrome, Firefox, Edge, Safari)

3. THEORETICAL BACKGROUND

3.1 Kaggle as a Data Science Hub

Kaggle is an online community of data scientists and machine learning practitioners. It provides access to public datasets, cloud-based Jupyter Notebook environments (Kaggle Kernels), and competitive data science challenges. Acquiring datasets from Kaggle is a standard practice for real-world data analysis and experimentation.

3.2 Version Control & Project Hosting via GitHub

GitHub is a cloud platform for software development and version control using Git. It allows researchers and developers to host code, track historical project revisions, collaborate asynchronously, and showcase project portfolios through structured repositories.

4. DETAILED EXPERIMENTAL PROCEDURE

Part A: Platform Account Setup

1. **Navigate to Kaggle & Register:**
   a. Open your web browser and go to `https://www.kaggle.com`.
   b. Click on the **Register** button in the top right corner.
   c. Sign up using your Google account or email address, choose a unique username, and set up your password.
2. **Navigate to GitHub & Sign Up:**
   a. Open a new browser tab and visit `https://github.com`.
   b. Click on **Sign up**.
   c. Enter your active email address, create a strong password, choose a unique username, and select email preferences.
3. **Complete Profile Verification:**
   a. Check your email inbox for verification links sent by Kaggle and GitHub.
   b. Click the respective verification links to activate full platform features (such as dataset downloading and code pushing).

Part B: Dataset Acquisition from Kaggle

1. **Log in and Access Datasets:**
   a. Log into your newly created Kaggle account.
   b. Click on the **Datasets** tab located in the left navigation panel.
2. **Search and Filter Datasets:**
   a. In the search bar, type a topic or dataset name (e.g., *Titanic - Machine Learning from Disaster*, *Iris Dataset*, or *World Happiness Report*).
   b. Use the filters on Kaggle (such as File Type: CSV, Data Usability Rating) to select a clean dataset.
3. **Download Dataset:**
   a. Click on the selected dataset to open its details page.
   b. Locate and click the **Download** icon/button (top-right of the dataset panel) to save the zip file or CSV file to your local computer.
4. **Extract and Organize Files:**
   a. Navigate to your local Downloads folder.
   b. Extract/unzip the contents into your project folder directory (e.g., `ds_lab_exp1`).

Part C: Performing Exploratory Data Analysis (EDA)

1. **Environment Setup:**
   a. Launch VS Code, Jupyter Notebook, or open Google Colab in your browser.
   b. Create a new notebook file named `analysis.ipynb` or a script `analysis.py` in your project directory.
2. **Load Dataset using Pandas:**
   a. Import the `pandas` library in Python.
   b. Use `pd.read_csv('your_dataset.csv')` to read the dataset into a pandas Dataframe.
3. **Perform Preliminary Analysis:**
   a. Execute `df.head()` to inspect the first 5 rows of data.
   b. Execute `df.info()` to inspect column data types, total non-null entries, and memory usage.
   c. Execute `df.describe()` to output summary statistics (mean, std, min, max, quantiles) for numeric columns.
   d. Execute `df.isnull().sum()` to check for missing values across all features.
4. **Save Script and Artifacts:**
   a. Save all code cells/script changes locally to ensure complete reproducibility.

Part D: Uploading Analysis to GitHub Repository

1. **Create a New Remote Repository:**
    a. Log into your GitHub account and click the **"+"** menu in the upper-right corner > **New repository**.
    b. Set the repository name to `data-science-lab-exp01`.
2. **Configure Repository Visibility:**
    a. Select **Public** so your submission can be viewed and evaluated.
3. **Initialize Repository Files:**
    a. Check the box **Add a README file**.
    b. In the **Add .gitignore** dropdown, select the **Python** template to ignore temporary execution files.
    c. Click **Create repository**.
4. **Upload Project Files:**
    a. On the newly created repository main page, click **Add file** > **Upload files**.
    b. Drag and drop your dataset file, `analysis.ipynb` (or `analysis.py`), and any accompanying document/notes.
5. **Update README Documentation:**
    a. Edit the `README.md` file to include a brief description of the experiment, dataset source URL, and a summary of EDA key observations.
6. **Commit and Push Changes:**
    a. Type a concise commit message in the "Commit changes" section (e.g., *"Upload Kaggle dataset analysis for Exp 01"*).
    b. Click **Commit changes** to push the dataset and script to main branch.

5. PYTHON EDA CODE TEMPLATE

Use the following snippet for your preliminary analysis in Part C:

```
import pandas as pd

# 1. Load the dataset downloaded from Kaggle
df = pd.read_csv('dataset.csv')

# 2. Display basic dataset details
print("--- Dataset Head ---")
print(df.head())

# 3. Structural Information
print("
--- Dataset Info ---")
print(df.info())

# 4. Summary Statistics
print("
--- Summary Statistics ---")
print(df.describe())

# 5. Missing Values Count
print("
--- Missing Values Count ---")
print(df.isnull().sum())
```